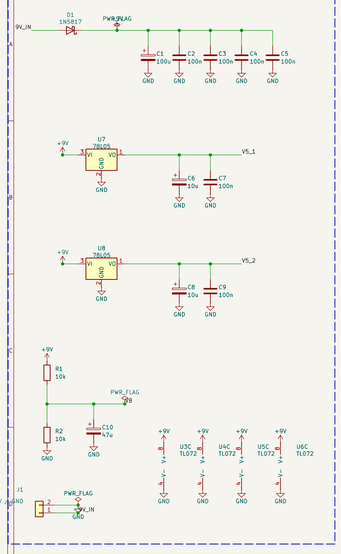
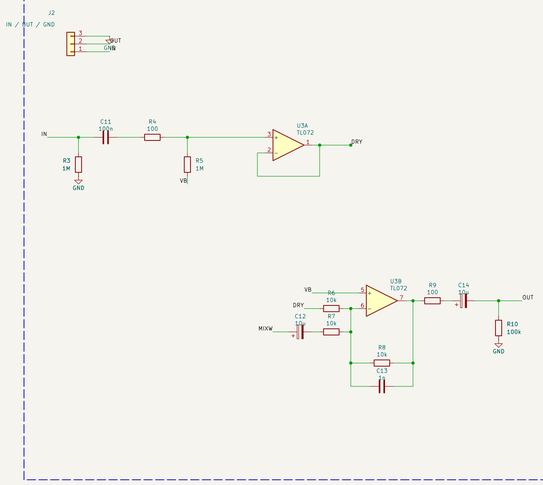
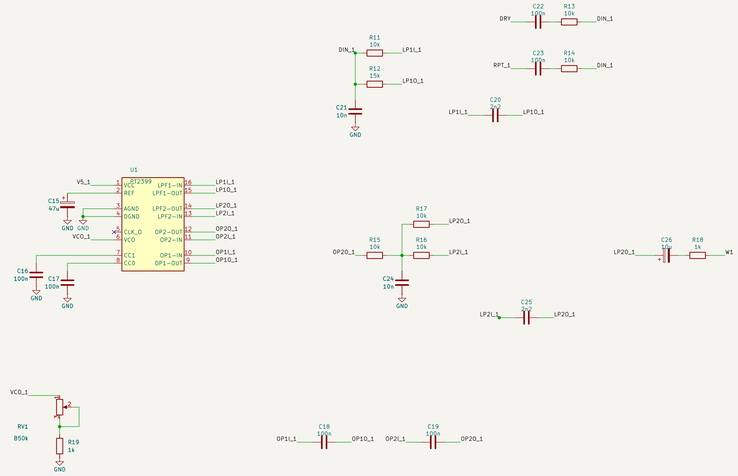
POWER: 9 V, VB 4.5 V for the TL072s, one 78L05 per PT2399



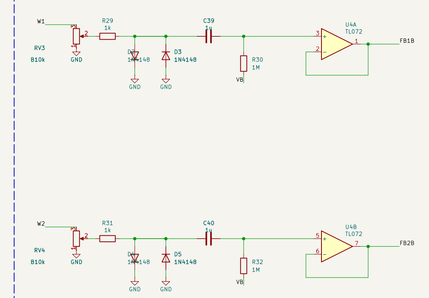
INPUT BUFFER (U3A) + OUTPUT MIXER (U3B): as Loop Echo / Small Time



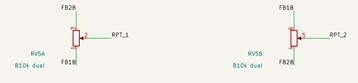
DELAY LINE 1: PT2399 (U1), network as Loop Echo / Small Time. TIME 1 = 1k + B50k



FEEDBACK (FB 1 / FB 2) -> limiter -> U4 buffers -> TWIST (dual gang): CCW = parallel, CW = Moeblus



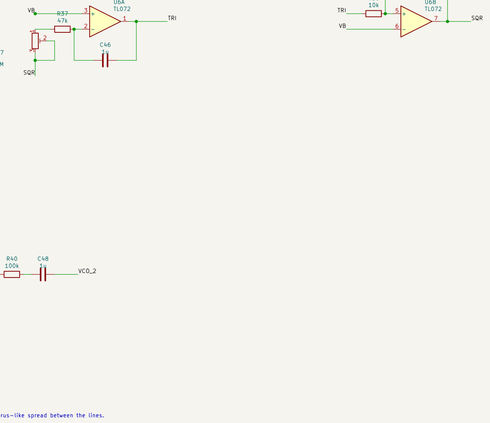
Limiter: the loop gain can reach 1 with FB up (and stays 1 across TWIST), so the 1N4148s cap self-oscillation at about +-0.6 V.



TWIST CCW: line 1 repeats itself, line 2 itself (parallel). CW: each line repeats the other (Moeblus: echoes at t1, t2, t1+t2).

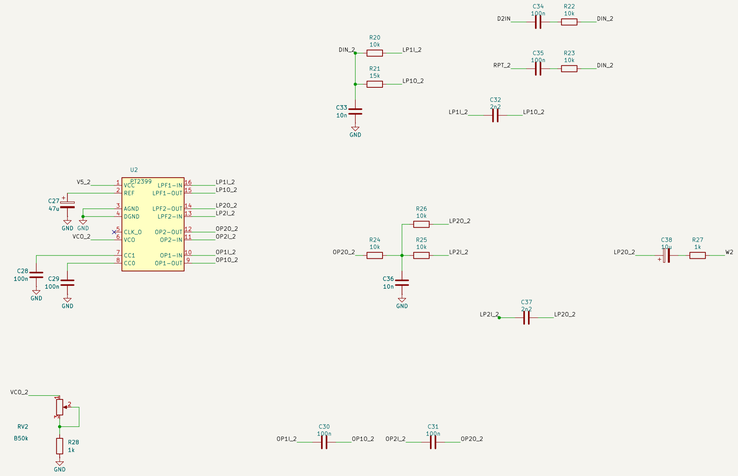
Middle: both. Fully up with both FB = same total loop gain at every TWIST setting.

MODULATION: LFO (U6) -> DEPTH trim -> line 2 VCO only. RATE / DEPTH are internal trims



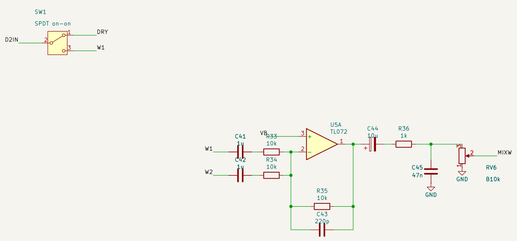
~0.5–12 Hz. Set RATE slow and DEPTH low for a chorus-like spread between the lines.

DELAY LINE 2: PT2399 (U2), network as Loop Echo / Small Time. TIME 2 = 1k + B50k



Line 2 input: DRY (parallel) or W1 (series), set by SW1 SERIES. Line 2 VCO also gets the LFO (chorus-like spread).

SERIES (SW1) + WET SUM (U5A) -> MIX. U5B spare (future stereo R)



Panel (1590BB): TIME 1, TIME 2, FB 1, FB 2, TWIST, MIX + SERIES toggle. Internal: RATE, DEPTH.

Blocks from Loop Echo (PT2399 network from Small Time); LFO on line 2
Two PT2399 lines, TWIST blends self / cross feedback (parallel .. Moeblus), SERIES toggle

Super-Freq

Sheet: /
File: DualDelay.kicad_sch

Title: Dual Delay (breadboard)

Size: A1	Date:	Rev: 0.1
KiCad E.D.A. 10.0.4		Id: 1/1